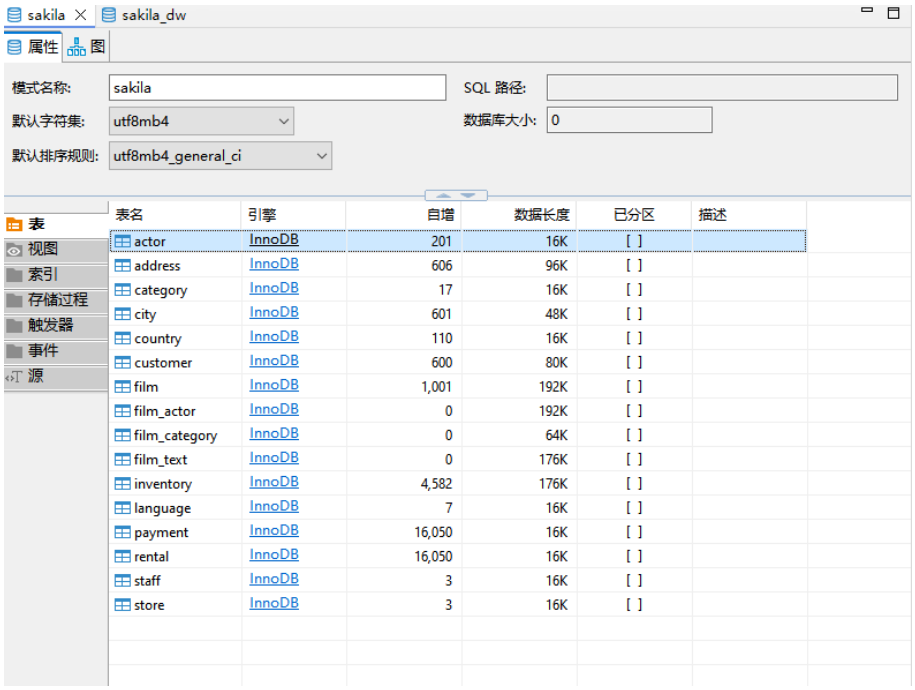
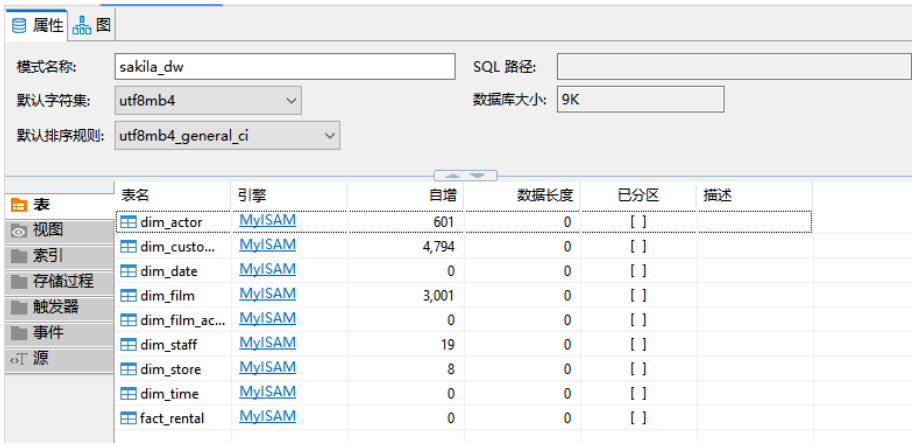
数据挖掘HW2报告

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1 数据准备

如下图，运行sql脚本后成功导入数据：





2 数据抽取

2.1 日期维度表

如下图是部分代码截图，为每个日期生成对应字段的数据。

```
# 计算日期属性
date_key = int(current_date.strftime("%Y%m%d"))
date_value = current_date
date_short = current_date.strftime("%d/%m/%Y")
date_medium = current_date.strftime("%b %d, %Y")
date_long = current_date.strftime("%B %d, %Y")
date_full = current_date.strftime("%A, %B %d, %Y")
day_in_year = current_date.timetuple().tm_yday
day_in_month = current_date.day
day_abbreviation = current_date.strftime("%a")
day_name = current_date.strftime("%A")
```

下图为导入后部分数据，其中可能有部分字段格式和ppt中不完全一致，个人认为这部分差异不是数据ETL过程的重点（因为不同的程序有不同的处理方式），因此不作处理，后续所有数据表都如此，不再赘述。

	date_key	date_value	date_short	date_medium	date_long	date_full
1	20,000,101	2000-01-01	01/01/2000	Jan 01, 2000	January 01, 2000	Saturday, January 01, 2000
2	20,000,102	2000-01-02	02/01/2000	Jan 02, 2000	January 02, 2000	Sunday, January 02, 2000
3	20,000,103	2000-01-03	03/01/2000	Jan 03, 2000	January 03, 2000	Monday, January 03, 2000
4	20,000,104	2000-01-04	04/01/2000	Jan 04, 2000	January 04, 2000	Tuesday, January 04, 2000
5	20,000,105	2000-01-05	05/01/2000	Jan 05, 2000	January 05, 2000	Wednesday, January 05, 2000

2.2 时间维度表

如图为生成一天内时间数据的代码：

```
for hour in range(24):
    for minute in range(60):
        for second in range(60):
            time_key = hour * 10000 + minute * 100 + second
            time_value = f"{hour:02d}:{minute:02d}:{second:02d}"
            hour24 = hour
            hour12 = hour % 12 if hour % 12 != 0 else 12
            am_pm = "AM" if hour < 12 else "PM"

            cursor.execute(
                """
                INSERT INTO dim_time VALUES (
                    %s, %s, %s, %s, %s, %s, %s
                )
            """,
                (time_key, time_value, hour24, hour12, minute, second, am_pm),
            )
```

如图为导入的部分数据：

	time_key	time_value	hours24	hours12	minutes	seconds	am_pm
1	0	00:00:00	0	12	0	0	AM
2	1	00:00:01	0	12	0	1	AM
3	2	00:00:02	0	12	0	2	AM
4	3	00:00:03	0	12	0	3	AM
5	4	00:00:04	0	12	0	4	AM
6	5	00:00:05	0	12	0	5	AM

2.3 员工维度表

下图为从staff表转换生成维度表数据的代码：

```
for staff in staffs:
    dw_cursor.execute(
        """
        INSERT INTO dim_staff (
            staff_last_update, staff_first_name, staff_last_name,
            staff_id, staff_store_id, staff_version_number,
            staff_valid_from, staff_valid_through, staff_active
        ) VALUES (%s, %s, %s, %s, %s, %s, %s, %s, %s)
    """,
        (
            staff["last_update"],
            staff["first_name"],
            staff["last_name"],
            staff["staff_id"],
            staff["store_id"],
            1,
            "1970-01-01",
            "2200-01-01",
            "Yes" if staff["active"] else "No",
        ),
    )
```

下图为导入后的数据：

		staff_last_up	s	staff_l				staff_va	staff_	staff_
1	19	2006-02-15 03:57:16	Mike	Hillyer	1	1	1	1970-01-01	2200-01-01	Yes
2	20	2006-02-15 03:57:16	Jon	Stephens	2	2	1	1970-01-01	2200-01-01	Yes

其中清空表会导致主键自增值重置，因此我在代码里显式设置了自增值为19。

2.4 用户维度表

下图为从各个外键关联的表中获取所需数据并进行join的代码：

```
cursor.execute(
    """
    SELECT
        c.customer_id, c.store_id, c.first_name, c.last_name, c.email,
        c.active, c.create_date, c.last_update,
        a.address, a.address2, a.district, a.postal_code, a.phone,
        ci.city,
        co.country
    FROM customer c
    JOIN address a ON c.address_id = a.address_id
    JOIN city ci ON a.city_id = ci.city_id
    JOIN country co ON ci.country_id = co.country_id
"""
)
```

下图为导入后的部分数据：

	customer_l		custo	cus	customer_email		cu	customer_address
1	4,794:2006-02-15 04:57:20	1	MARY	SMITH	MARY.SMITH@sakilacustomer.org	Yes	2006-02-14	1913 Hanoi Way
2	4,795:2006-02-15 04:57:20	2	PATRICIA	JOHNSON	PATRICIA.JOHNSON@sakilacustomer.org	Yes	2006-02-14	1121 Loja Avenue
3	4,796:2006-02-15 04:57:20	3	LINDA	WILLIAMS	LINDA.WILLIAMS@sakilacustomer.org	Yes	2006-02-14	692 Joliet Street
4	4,797:2006-02-15 04:57:20	4	BARBARA	JONES	BARBARA.JONES@sakilacustomer.org	Yes	2006-02-14	1566 Inegl Manor
5	4,798 2006-02-15 04:57:20	5	ELIZABETH	BROWN	ELIZABETH.BROWN@sakilacustomer.org	Yes	2006-02-14	53 Idfu Parkway

2.5 商店维度表

下图为从外键关联表提取数据的代码：

```
cursor.execute(
    """
    SELECT
        s.*,
        a.address,
        a.district,
        a.postal_code,
        a.phone as phone_number,
        ci.city,
        co.country,
        st.first_name as manager_first_name,
        st.last_name as manager_last_name,
        st.staff_id as manager_staff_id
    FROM store s
    JOIN address a ON s.address_id = a.address_id
    JOIN city ci ON a.city_id = ci.city_id
    JOIN country co ON ci.country_id = co.country_id
    JOIN staff st ON s.manager_staff_id = st.staff_id
"""
)
```

下图为导入的数据：

		store_last_up		store_addr				store_	stc			st		store_
1	8	2006-02-15 04:57:12	1	47 MySakila Drive	Alberta			Lethbridge	Canada	1	Mike	Hillyer	1	1970
2	9	2006-02-15 04:57:12	2	28 MySQL Boulevar	QLD			Woodridge	Australia	2	Jon	Stephens	1	1970

2.6 演员维度表

如下图为代码：

```
for actor in actors:
    dw_cursor.execute(
        """
        INSERT INTO dim_actor (
            actor_last_update, actor_last_name, actor_first_name, actor_id
        ) VALUES (%s, %s, %s, %s)
    """,
        (
            actor["last_update"],
            actor["last_name"],
            actor["first_name"],
            actor["actor_id"],
        ),
    )
```

下图为部分数据：

actor_key	actor_last_update	actor_last_name	actor_first_name	actor_id
601	2006-02-15 04:34:33	GUINESS	PENELOPE	1
602	2006-02-15 04:34:33	WAHLBERG	NICK	2
603	2006-02-15 04:34:33	CHASE	ED	3
604	2006-02-15 04:34:33	DAVIS	JENNIFER	4
605	2006-02-15 04:34:33	LOLLOBRIGIDA	JOHNNY	5

2.7 电影维度表

下图为获取film数据的代码：

```
cursor.execute(
    """
    SELECT
        f.*,
        l.name as language_name,
        IFNULL(ol.name, 'Not Applicable') as original_language_name,
        COUNT(DISTINCT fa.actor_id) as actor_count
    FROM film f
    JOIN language l ON f.language_id = l.language_id
    LEFT JOIN language ol ON f.original_language_id = ol.language_id
    LEFT JOIN film_actor fa ON f.film_id = fa.film_id
    GROUP BY f.film_id
"""
)
films = cursor.fetchall()
```

下图为dim_film表的部分数据：

		film_last_up	film_title	film		f										
1	3,001	2006-02-15 05:03:42	ACADEMY DINOSAUR	A Epic Dram	2,006	English	Not A	0.99	6	86	20.99	PG	Parenta	No	No	Yes
2	3,002	2006-02-15 05:03:42	ACE GOLDFINGER	A Astoundin	2,006	English	Not A	4.99	3	48	12.99	G	General	Yes	No	Yes
3	3,003	2006-02-15 05:03:42	ADAPTATION HOLES	A Astoundin	2,006	English	Not A	2.99	7	50	18.99	NC-17	No One	Yes	No	Yes
4	3,004	2006-02-15 05:03:42	AFFAIR PREJUDICE	A Fanciful D	2,006	English	Not A	2.99	5	117	26.99	G	General	No	Yes	No
5	3,005	2006-02-15 05:03:42	AFRICAN EGG	A Fast-Pace	2,006	English	Not A	2.99	6	130	22.99	G	General	No	No	Yes

下图为提取film_actor表数据并计算权重因子的代码：

```python
for fa in film_actors:
    # 获取电影和演员的key
    dw_cursor.execute(
        "SELECT film_key FROM dim_film WHERE film_id = %s", (fa["film_id"],)
    )
    film_key = dw_cursor.fetchone()[0]

    dw_cursor.execute(
        "SELECT actor_key FROM dim_actor WHERE actor_id = %s", (fa["actor_id"],)
    )
    actor_key = dw_cursor.fetchone()[0]

    # 计算权重因子 (1/演员数量)
    actor_count = film_actor_counts.get(fa["film_id"], 1)
    weighting_factor = 1.0 / actor_count if actor_count > 0 else 0

    dw_cursor.execute(
        """
        INSERT INTO dim_film_actor_bridge (film_key, actor_key, actor_weighting_factor)
        VALUES (%s, %s, %s)
        """,
        (film_key, actor_key, weighting_factor),
    )
```

下图为dim_film_actor_bridge表的部分数据：

film_key	actor_key	actor_weighting_factor
3,001	601	0.1
3,001	610	0.1
3,001	620	0.1
3,001	630	0.1
3,001	640	0.1
3,001	653	0.1
3,001	708	0.1
3,001	762	0.1
3,001	788	0.1
3,001	798	0.1
3,002	619	0.25
3,002	685	0.25
3,002	690	0.25
3,002	760	0.25

其中权重因子和ppt上演示的并不一致，本人查看了film_actor表数据，确认了数据的正确性，或许是ppt上演示的数据有误。

2.8 租赁事实表

下图为获取外键关联表数据的代码：

```python
cursor.execute(
    """
    SELECT r.*, i.film_id, i.store_id, p.amount,
           TIMESTAMPDIFF(SECOND, r.rental_date, r.return_date) AS duration_seconds
    FROM rental r
    JOIN inventory i ON r.inventory_id = i.inventory_id
    LEFT JOIN payment p ON r.rental_id = p.rental_id
    WHERE r.return_date IS NOT NULL
"""
)
```

下图为部分导入数据：

	123	123	123	123	123 rei	123 reti	123 r	123	123	123 ren	rental_last_upc	123
1	4,923	19	3,080	8	20,050,524	20,050,526	225,330	1	1	169,860	2006-02-15 21:30:53	1
2	5,252	19	3,333	9	20,050,524	20,050,528	225,433	1	1	333,960	2006-02-15 21:30:53	2
3	5,201	19	3,373	9	20,050,524	20,050,601	230,339	1	1	688,140	2006-02-15 21:30:53	3
4	5,126	20	3,535	8	20,050,524	20,050,603	230,441	1	1	787,140	2006-02-15 21:30:53	4
5	5,015	19	3,450	9	20,050,524	20,050,602	230,521	1	1	710,880	2006-02-15 21:30:53	5

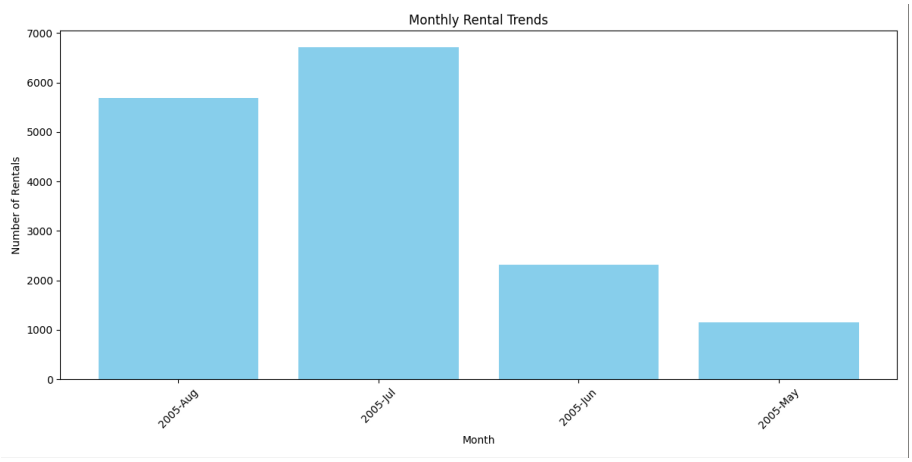
3 OLAP操作

3.1 下钻

下图为代码：

```
cursor.execute(
    """
    SELECT
        d.year_month_abbreviation AS month,
        COUNT(*) AS rental_count
    FROM
        fact_rental f
    JOIN
        dim_date d ON f.rental_date_key = d.date_key
    GROUP BY
        d.year_month_abbreviation
    ORDER BY
        d.year_month_abbreviation;
    """
)
```

逻辑为从租赁事实表中按照月份分类钻取，并展示每个月份的租赁量，下图为可视化：



3.2 切片

下图为代码：

```
cursor.execute(
    f"""
    SELECT
        CONCAT(dd.month_name, ' ', dd.year4) AS month_year,
        COUNT(*) AS rental_count
    FROM
        fact_rental fr
    JOIN
        dim_film df ON fr.film_key = df.film_key
    JOIN
        dim_date dd ON fr.rental_date_key = dd.date_key
    WHERE
        df.film_in_category_{category.lower()} = 'Yes'
    GROUP BY
        dd.year4, dd.month_name, dd.month_number
    ORDER BY
        dd.year4, dd.month_number;
    """
)
```

逻辑为选中租赁事实表中特定电影类别的数据进行统计，可视化如下图（类别为'Action'）：

